Logistic Regression

Thursday, October 8, 2026

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Quantitative Reasoning II · INF 1220 · University of Austin

Lesson 11

Yes-or-No Outcomes

- Field goal attempts: made or missed
- Loans: repaid or defaulted
- Hospital patients: readmitted within 30 days or not
- College students: graduated in four years or not
- Incumbents: reelected or defeated

In each case we want to know how the **probability** of an event occurring changes with X.

NFL Field Goal Attempts

- NFL field goal attempts from 2001–2005 and 2021–2025
- Regular season and playoffs: 10,503 kicks
- Each row is one kick

The two blocks are 20 years apart, so we can compare kickers then and now.

Made: 1 if the kick went through, 0 if it missed. A blocked kick counts as a miss.

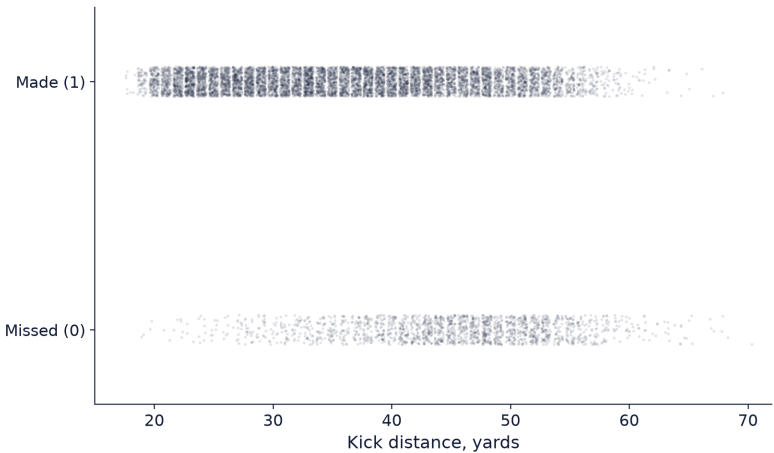
Distance: the official field goal distance in yards, from where the ball is kicked to the goalposts.

nflverse play-by-play data, pulled October 7, 2026.

Data Snapshot

Kicker	Team	Game	Distance	Made
Adam Vinatieri	Patriots	2001 playoffs, in the snow	45	1
Adam Vinatieri	Patriots	Super Bowl, 2001 season	48	1
Adam Vinatieri	Patriots	Super Bowl, 2003 season	31	0
Justin Tucker	Ravens	2021, at Detroit	66	1
Brandon Aubrey	Cowboys	2024, at Carolina	70	0
Cam Little	Jaguars	2025, at Las Vegas	68	1

Made or Missed, by Distance



Share Made by Distance

Distance, yards	Kicks	Made	Share made
18–29	2,673	2,582	96.6%
30–39	3,030	2,688	88.7%
40–49	3,126	2,293	73.4%
50–59	1,593	1,049	65.9%
60 and over	81	30	37.0%

How does the probability of making a kick change with distance?

One Option: Average at Each Distance

To predict the probability at a distance, use the share of kicks made from that distance. For a 50-yard kick:

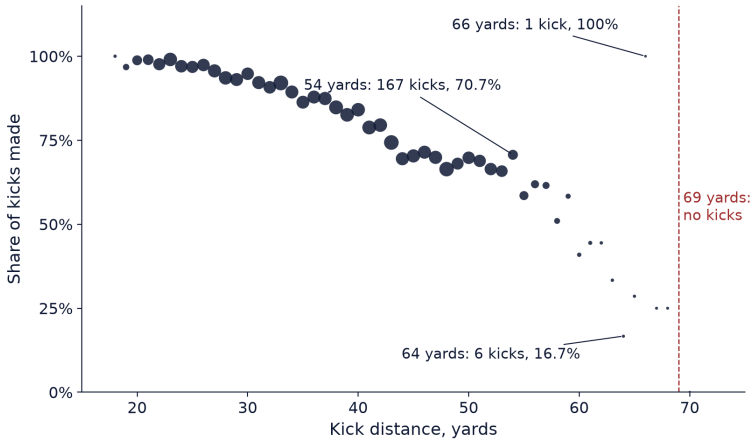
```
# Keep only the 50-yard kicks
kicks_50 = kicks[kicks["kick_distance"] == 50]

print("kicks:", len(kicks_50))
print("made: ", kicks_50["made"].sum())
print("share:", round(kicks_50["made"].mean(), 3))
```

```
kicks:  268
made:  187
share:  0.698
```

187 of 268 kicks were made: $187/268 = 69.8\%$.

Option 1: Average at Each Distance



Larger dots are distances with more kicks.

Problem 1: Few Kicks, Jumpy Averages

Yards	Kicks	Made	Share
50	268	187	69.8%
51	260	179	68.8%
52	262	174	66.4%
53	231	152	65.8%
54	167	118	70.7%
55	135	79	58.5%

Yards	Kicks	Made	Share
63	9	3	33.3%
64	6	1	16.7%
65	7	2	28.6%
66	1	1	100.0%
67	4	1	25.0%
68	4	1	25.0%

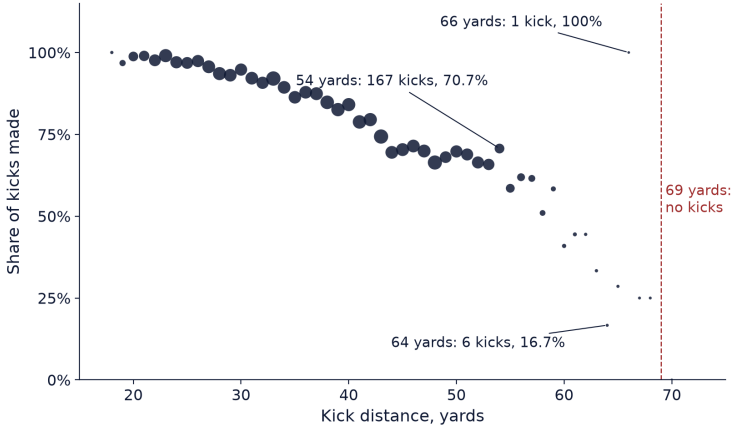
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Yards	Kicks	Made	Share
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64	**6**	**1**	**16.7%**
65	**7**	**2**	**28.6%**
66	**1**	**1**	**100.0%**
67	4	1	25.0%
68	4	1	25.0%

Problem 2: No Kicks, No Prediction

What about a 69-yard kick?



Problem 3: More Predictors means Smaller Groups

Distance is not the only thing that could affect a kick:

- **Era:** kickers in 2021–2025 may be better than kickers in 2001–2005
- **Indoors or outdoors:** a stadium with a dome or a closed roof has no wind, rain, or snow

To use averages, we would average separately within each group at each distance: 60-yard kicks outdoors in 2001–2005, 60-yard kicks indoors in 2001–2005, and so on.

Calculating an average for every group is more work, and more of the groups have few kicks or none.

Option 2: Least Squares on a 0/1 Outcome

Another option is to run an ordinary least squares regression:

$$\widehat{\text{made}} = 1.276 - 0.0119 \times \text{distance}$$

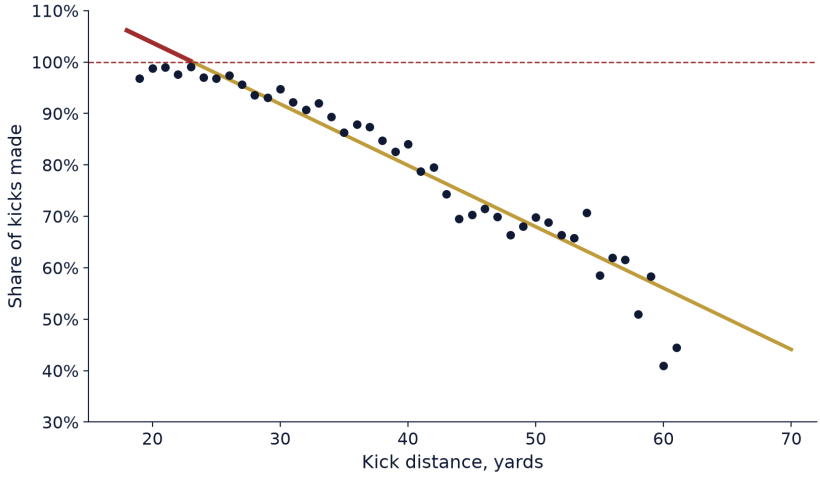
This is called the **linear probability model**.

- With a 0/1 outcome, $\widehat{\text{made}}$ is the **predicted probability** that a kick at that distance is made.

Interpretation:

- **Slope:** each additional yard is associated with a 1.19-percentage-point lower probability of making the kick, on average.

Linear Probability Model



Limitations of the Linear Probability Model

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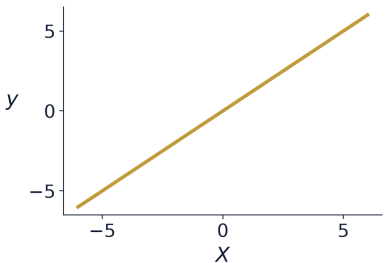
What is a 106% chance of making a kick?

- Every kick shorter than 24 yards has a predicted probability above 100%: 990 kicks, 9.4% of the data
- From 60 yards and over, the line predicts 53% on average; 37% went through
- Past 107 yards the line predicts a negative probability

What If We Bent the Edges?

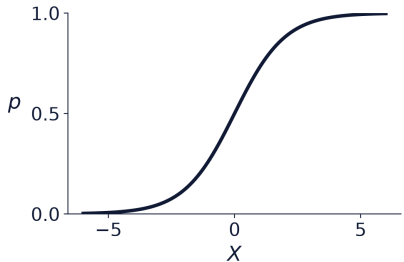
Line

$$y = \beta_0 + \beta_1 X$$



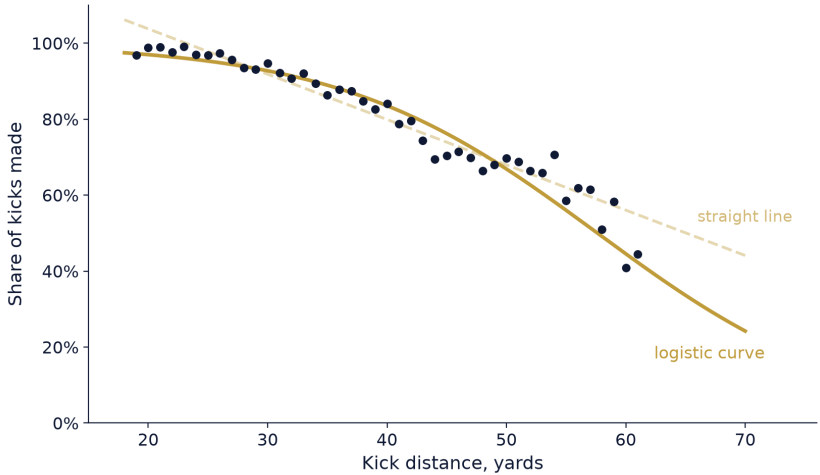
Logistic curve

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X$$



The right-hand side is the same line. On the left, $\ln\big(p/(1-p)\big)$ bends it so that p stays between 0 and 1.

Logistic Regression Fits This Curve



Odds

The **odds** of an event are its probability divided by the probability it does not happen:

$$\text{odds} = \frac{p}{1-p}$$

For kicks, the odds are makes per miss:

Distance, yards	Made	Missed	Share made	Odds
18–29	2,582	91	96.6%	28.4
30–39	2,688	342	88.7%	7.9
40–49	2,293	833	73.4%	2.8
50–59	1,049	544	65.9%	1.9
60 and over	30	51	37.0%	0.6

Odds of 1 are one make per miss: a 50% chance.

Log Odds

The **log odds** are the natural log of the odds: $\ln\left(\frac{p}{1-p}\right)$.

Probability	Odds	Log odds
25%	0.33	−1.10
50%	1	0
75%	3	1.10
90%	9	2.20
99%	99	4.60

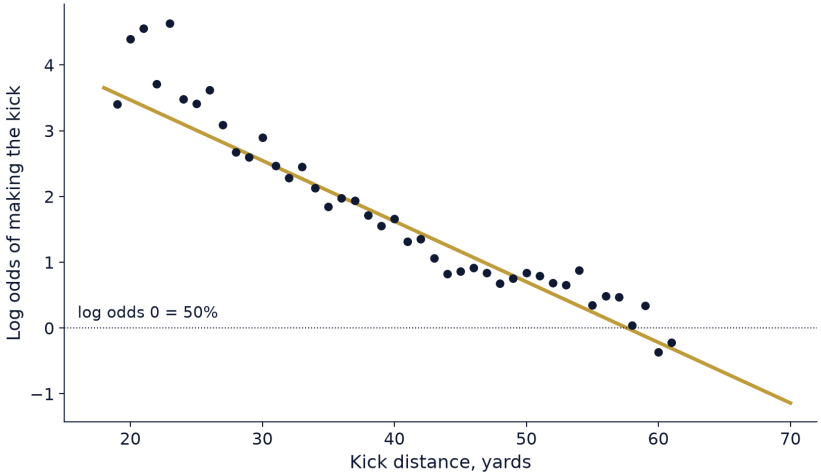
Log odds can be any number. A 50% chance is log odds of 0.

Logistic Regression Model

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \times \text{distance}$$

p is the probability that a kick is made.

Log Odds by Distance with Logistic Regression Line



Logistic Regression: Software Output

```
==========================================================
Dep. variable: made                   Observations    10503
Model: Logit
----------------------------------------------------------
                            coef   std err        z    P>|z|
----------------------------------------------------------
const                     5.3108    0.1331    39.92    0.000
kick_distance            -0.0922    0.0030   -30.80    0.000
==========================================================
```

The coefficients are the intercept and slope of the log-odds line:

$$\ln\left(\frac{p}{1-p}\right) = 5.3108 - 0.0922 \times \text{distance}$$

The z-statistic is the coefficient divided by its standard error, the same as t in L10.

Making a Prediction

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$$2.01 = \frac{p}{1-p} \quad \Rightarrow \quad p = \frac{2.01}{3.01} = 0.668$$

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The model predicts that about 67% of 50-yard kicks go through.

Predicted Probabilities

Distance, yards	Predicted probability	Change from 10 yards closer
20	97.0%	
30	92.7%	−4.3
40	83.5%	−9.2
50	66.9%	−16.6
60	44.5%	−22.4

- The change in probability is nonlinear: probability decreases by 4.3 percentage points from 20–30 yards, but 22.4 percentage points between 50 and 60 yards.
- A LPM predicts a constant change of 11.9 percentage points per 10 yards regardless of where we are on the field.

Interpreting the Coefficient

Sign: the distance coefficient is negative, so the probability of making a kick falls as distance rises.

Size: -0.0922 is the change in the log odds for each additional yard.

- $e^{-0.0922} = 0.91$ odds so each yard multiplies the odds of making by $0.91 \Rightarrow$ .91-1 = -0.09 $\Rightarrow$ 9% decrease in the odds of making the kick per yard.

There is no single change in probability because the change in probability depends on our starting point. The log odds change is constant, but the probability change is not.

Have Kickers Improved?

	2001–2005	2021–2025
Kicks	4,909	5,594
Share made	78.9%	85.3%
Average distance, yards	36.3	39.5
Share of kicks from 50 yards or more	9.1%	21.9%

Kickers in 2021–2025 made 6.4 percentage points more of their kicks.

How good is this as an estimate of the improvement over time?

Adding a Second Predictor

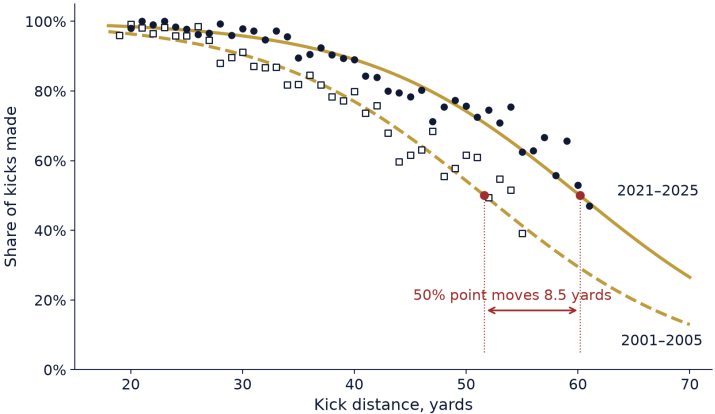
$$\text{recent} = \begin{cases} 1 & \text{if the kick was in 2021–2025} \\ 0 & \text{if it was in 2001–2005} \end{cases}$$

Outcome: made, 0/1	**(1)**	**(2)**
Distance, yards	-0.092^{***}	-0.104^{***}
Recent, 0/1		0.886^{***}
Constant	5.311^{***}	5.358^{***}
n	10,503	10,503

$^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.001$

Column (2) shows us that holding distance constant, the chance of making a kick is higher in 2021–2025 than in 2001–2005.

Specialized Curve for Each Category



How does this affect our estimate of improvement over time?

- Remember, our original estimate was a difference in the share of kicks made as estimated by the raw data: an increase of 6.4 percentage points.
- Now, we use model (2) to estimate a 50-yard kick in each era (holding distance constant):

2001–2005	54.2%
2021–2025	74.2%

- Our logistic model estimates the improvement at 50 yards to be 20 percentage points, not 6.4!

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Why?

- In 2021–2025, there are way more kick attempts at 50+ yards: 21.9% of all attempts, vs 9.1% in 2001–2005. The raw comparison doesn't account for the different distributions of kicks by distance.

Practice

Indoor is 1 if the stadium had a dome or a closed roof, and 0 if not

Outcome: made, 0/1	**Coef.**	**Std. Err.**	***p*-value**
Constant	5.346	0.138	<0.001
Distance, yards	-0.105	0.003	<0.001
Recent, 0/1	0.843	0.058	<0.001
Indoor, 0/1	0.365	0.066	<0.001

1. Interpret the sign of the indoor coefficient.
2. Predict the probability of making a 50-yard kick indoors in 2021–2025.

Practice

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Constant	5.346	0.138	<0.001
Distance, yards	-0.105	0.003	<0.001
Recent, 0/1	0.843	0.058	<0.001
Indoor, 0/1	0.365	0.066	<0.001

Sign: positive, so at the same distance and era indoor kicks are more likely to go through. $e^{0.365} = 1.44$, so a dome raises the odds of making by 44%.

Prediction: $5.346 - 0.105 \times 50 + 0.843 + 0.365 = 1.30$, and $e^{1.30}/(1 + e^{1.30}) = 79\%$
The same kick outdoors: 72%